TEXORA · LECTURE SHEET

Ordinary Differential Equations

Higher-order linear equations with constant coefficients — complementary function, particular integrals by operator methods and by variation of parameters — Cauchy's and Legendre's equations, and simultaneous first-order systems.

MATH-1201 · MATHEMATICS-II · TOPIC 2 · LEVEL 1 TERM 2

1. The framework

With D = d/dx, a linear ODE with constant coefficients is f(D)y = Q(x). Its general solution is

$$y = C.F. + P.I.$$

the **complementary function** (the general solution of f(D)y = 0, carrying all the arbitrary constants) plus any one **particular integral** of the full equation. The whole topic is a set of reliable machines for these two parts.

2. The complementary function: auxiliary equation

Try y = e^{mx} in f(D)y = 0: the **auxiliary equation** f(m) = 0 decides everything.

ROOTS OF F(M) = 0	C.F. CONTRIBUTION
Real, distinct: m_1, m_2	$Ae^{m_1x} + Be^{m_2x}$
Real, repeated: m, m	$(A + Bx)e^{mx}$
Complex: $\alpha \pm i\beta$	$e^{\alpha x}(A \cos \beta x + B \sin \beta x)$
Repeated complex pair	$e^{\alpha x}[(A + Bx) \cos \beta x + (C + Dx) \sin \beta x]$

WORKED EXAMPLES

$(D^2 - 3D + 2)y = 0$: m = 1, 2 → $y = Ae^x + Be^{2x}$.

$(D^2 + 2D + 1)y = 0$: m = −1, −1 → $y = (A + Bx)e^{-x}$.

$(D^2 + 4D + 13)y = 0$: m = −2 ± 3i → $y = e^{-2x}(A \cos 3x + B \sin 3x)$ — damped oscillation, the shape of every decaying circuit and spring.

3. Particular integrals: operator shortcuts

Write P.I. = (1/f(D)) Q(x) and use the standard rules:

- **Q = e^{ax}:** P.I. = e^{ax}/f(a), provided f(a) ≠ 0. If f(a) = 0 (the *failure case*), multiply by x and differentiate the denominator: P.I. = x e^{ax}/f′(a) (repeat as needed).
- **Q = sin ax or cos ax:** replace D^2 by $-a^2$ wherever it appears: P.I. = Q-evaluated with $f(D^2) \to f(-a^2)$, when the result is defined. If $f(-a^2) = 0$ (resonance), the answer gains a factor x: e.g. $(1/(D^2 + a^2))\cos ax = x \sin ax/(2a)$.
- **Q = x^n (polynomial):** expand 1/f(D) in ascending powers of D (binomial series) and apply term by term; the series terminates.
- **Q = e^{ax}V:** shift: (1/f(D)) e^{ax}V = e^{ax} (1/f(D + a)) V.
- **Q = xV:** (1/f(D)) xV = x (1/f(D)) V − (f′(D)/f(D)²) V.

WORKED EXAMPLES

$(D^2 - 1)y = e^x$: f(1) = 0, failure → P.I. = x e^x/f′(1) = x e^x/(2D)|_{D=1} = **x e^x/2**.

$(D^2 + 4)y = \cos 2x$: resonance → P.I. = **x sin 2x/4**.

$(D^2 - 3D + 2)y = x^2$: $1/f(D) = \tfrac{1}{2}(1 - (3D - D^2)/2)^{-1} = \tfrac{1}{2}[1 + (3D - D^2)/2 + \ldots]$; applying to x^2 gives P.I. = $\tfrac{1}{2}[x^2 + 3x + 7/2]$ = **$x^2/2 + 3x/2 + 7/4$**. (Check by substitution: $y''-3y'+2y$ with $y = x^2/2 + 3x/2 + 7/4$ gives $1 - 3x - 9/2 + x^2 + 3x + 7/2 = x^2$ ✓.)

4. Variation of parameters — the general machine

When Q(x) fits no shortcut (tan x, sec x, 1/x…), use the C.F. itself. If y = Au(x) + Bv(x) solves the homogeneous equation, seek a P.I. of the form y = A(x)u + B(x)v with A, B varying, imposing A′u + B′v = 0. Then, with W the Wronskian of u and v, for y″ + Py′ + Qy = R(x):

$$A' = -vR/W, \quad B' = uR/W, \quad W = uv' - u'v$$

WORKED EXAMPLE

$y'' + y = \sec x$. Here u = cos x, v = sin x, W = 1, R = sec x.

A′ = −sin x sec x = −tan x → A = ln|cos x|. B′ = cos x sec x = 1 → B = x.

P.I. = cos x · ln|cos x| + x sin x.

General solution: y = A cos x + B sin x + x sin x + cos x ln|cos x| — the classic result, obtainable no other elementary way.

5. Cauchy’s (Euler’s) equation

Equations with coefficients that are powers of x matching the derivative order — $x^2y'' + axy' + by = Q(x)$ — become constant-coefficient equations under **x = e^z** ($z = \ln x$). With $\theta = d/dz$: $xy' = \theta y$ and $x^2y'' = \theta(\theta - 1)y$.

WORKED EXAMPLES

$x^2y'' + 3xy' + y = 0$: indicial $m^2 + 2m + 1 = (m + 1)^2 = 0$ (try $y = x^m$, or transform) → repeated root -1 → $y =$ **(A + B ln x)/x**.

$x^2y'' - xy' + y = x$: homogeneous part $(m - 1)^2 = 0$ → $y_h = (A + B \ln x)x$. For the P.I., $z = \ln x$ gives $(\theta - 1)^2Y = e^z$, whose resonant P.I. is $z^2e^z/2$ → P.I. = **x(ln x)²/2**.

6. Legendre's linear equation

The same trick, shifted: $(a + bx)^n$-weighted equations — $(a + bx)^2y'' + p(a + bx)y' + qy = Q$ — become constant-coefficient under **a + bx = e^z**. Then $(a + bx)y' = b\theta y$ and $(a + bx)^2y'' = b\theta(b\theta - b)y$… in operator form, $(a+bx)D_x = b\theta$ and $(a+bx)^2D_x^2 = b^2\theta(\theta-1)$ — proceed exactly as for Cauchy.

WORKED EXAMPLE

$(1 + x)^2y'' + (1 + x)y' - y = 0$. With $1 + x = e^z$ ($b = 1$): $\theta(\theta - 1)Y + \theta Y - Y = (\theta^2 - 1)Y = 0$ → $\theta = \pm 1$ → $Y = Ae^z + Be^{-z}$ → $y =$ **A(1 + x) + B/(1 + x)**. Direct substitution confirms both terms.

7. Simultaneous first-order equations

For coupled systems (dx/dt, dy/dt with x, y mixed), eliminate: differentiate one equation and substitute from the other to get a single higher-order equation for one variable — or exploit symmetry by adding/subtracting the equations.

WORKED EXAMPLE

$dx/dt + y = e^t$ and $dy/dt + x = e^t$.

Add: $(x + y)' + (x + y) = 2e^t$ → $x + y = e^t + Ae^{-t}$.

Subtract: $(x - y)' - (x - y) = 0$ → $x - y = Be^t$.

Hence $x = \frac{1}{2}[(1 + B)e^t + Ae^{-t}]$, $y = \frac{1}{2}[(1 - B)e^t + Ae^{-t}]$ — two constants A, B for a system of total order two, as required.

8. Key points

- Every solution = C.F. (all constants, from the auxiliary equation) + P.I. (any one particular solution).

- Operator shortcuts cover e^{ax}, sin/cos, polynomials and their products; resonance (failure) always costs a factor x per repetition.
- Variation of parameters is the universal fallback — it needs only the C.F. and the Wronskian.
- Cauchy: x = e^z. Legendre: a + bx = e^z. Both are the same idea — make the coefficients constant by a logarithmic change of variable.
- Simultaneous systems: eliminate, or add/subtract when symmetric; the number of arbitrary constants equals the total order.

9. Practice

Answer in your own words; brief model answers follow.

1. Solve $(D^2 - 5D + 6)y = 0$.
2. Solve $(D^2 + 9)y = \sin 3x$ (mind the resonance).
3. Find the P.I. of $(D^2 - 4D + 4)y = e^{2x}$.
4. Solve $x^2y'' + xy' - y = 0$.
5. Use variation of parameters to solve $y'' + y = \tan x$.
6. Solve the system $dx/dt = y$, $dy/dt = -x$ with $x(0) = 1$, $y(0) = 0$, and interpret the motion.

Answer key

1. $m = 2, 3 \rightarrow y = Ae^{2x} + Be^{3x}$.
2. C.F. = $A\cos 3x + B\sin 3x$. Resonance: P.I. = $-x\cos 3x/(2\cdot 3) = -x\cos 3x/6$ (for $1/(D^2 + a^2))\sin ax$ the resonant P.I. is $-x\cos ax/(2a)$). General solution $y = A\cos 3x + B\sin 3x - (x\cos 3x)/6$.
3. $(D - 2)^2$: e^{2x} fails twice (double root $m = 2$): P.I. = $x^2e^{2x}/2! = x^2e^{2x}/2$.
4. Try $y = x^m$: $m(m - 1) + m - 1 = m^2 - 1 = 0 \rightarrow m = \pm 1 \rightarrow y = Ax + B/x$.
5. $u = \cos x$, $v = \sin x$, $W = 1$, $R = \tan x$. $A' = -\sin x\tan x = -(\sec x - \cos x)\cdot\ldots$ precisely $A' = -\sin^2 x/\cos x = \cos x - \sec x$, so $A = \sin x - \ln|\sec x + \tan x|$. $B' = \cos x\tan x = \sin x$, so $B = -\cos x$. P.I. = $\cos x\sin x - \cos x\ln|\sec x + \tan x| - \cos x\sin x = -\cos x\ln|\sec x + \tan x|$. General solution $y = A\cos x + B\sin x - \cos x\ln|\sec x + \tan x|$.
6. Eliminate: $x'' = y' = -x \rightarrow x'' + x = 0 \rightarrow x = \cos t$ (with the initial conditions), $y = x' = -\sin t$. The point (x, y) travels the unit circle clockwise — the system is uniform circular motion written as two ODEs.